expert opinion of top specialists who have vast experience in this field from our country. As a part of this effort, we have published 24 chapters in the latest issue of Indian Heart Journal covering every aspect of Lipid management.

Q. As per the expert group findings, what must be the Low-Density Lipoprotein (LDL) and non-High Density Lipoprotein (non-HDL) cholesterol goals for the Indian population?

Since a large population has high Triglycerides (>150 mg/dl), non-HDL is recommended as a co-primary goal along with LDL-cholesterol. And we recommended that every patient with dyslipidemia should be assessed for associated risk factors. We have categorised the goals as per the risk. A low to moderate risk population with one risk factor must keep LDL<100 mg/dl or non HDL <130 mg/dl. A high risk population with diabetes and hypertension and one risk factor must keep below 70 mg/dl.

Very high risk populations are those who are suffering from heart attack, angina, >50% block in 2 coronary arteries in CT angiography, stroke, limb vessel disease , diabetes of more than 20 years with complications, must keep it below <55 mg/dl and non HDL <85 mg/dl. Extremely high risk heart patients getting recurrent heart attack or stroke or limb disease and those cardiovascular disease patients with combined genetic factor of Familial Hypercholesterolemia and high Lp(a) target is to keep LDL-C below 40 mg/dl.

Q. What are the common drugs to treat lipid disorder in India?

Statins are the mainstay of the treatment. It acts on the liver to reduce cholesterol and it is a safe and effective drug used for more than three decades now. It reduces heart attacks, stroke and cardiovascular death.

Common side effects are myalgia (muscle pain)in less than 10% cases. Statins should be avoided in advanced liver disease patients. New onset of diabetes is seen in patients with higher dosage of statins and particularly patients with metabolic syndrome (central obesity).

Some of the non-statin drugs are Ezetimibe that prevent cholesterol absorption from intestine and reduces cholesterol by 10-15% and it is safe and better tolerated. Bempedoic acid (BA) 180 mg acts on the liver and reduces LDL-C by 15% and does not cause muscle pain. PCSK9 Inhibitors like Evolucumab subcutaneous injection twice a month is very effective and safe but a costly drug. Now a new drug Inclisiran is also available which reduces synthesis of PCSK9 and the advantage is that only 2 injections in a year are required and reduces cholesterol by 50%.

Q. Do you treat dyslipidemia in acute heart attack patients differently?

Yes, we know that heart attack patients have recurrence of attack very common in the first 3 months.

So, we are aggressive in lowering LDL cholesterol<55 mg within a few weeks. Initially we use high doses of statin (Atorvastatin 80 mg or rosuvastatin 40 mg) with ezetimibe and get repeat lipid profile at 4 weeks , if the goal is not reached we add Bempedoic acid and still goal is not attained then PCSK9I or Inclisiran is added.

For chronic stable angina patients we recommend high dosage of statin with ezetimibe (Rosuvastatin 20 mg or Atorvastatin 40mg with ezetimibe) to reach the goal of LDL-C<55 mg/dl. Combination is more effective and safe. If statin not tolerated then oral non statin drugs to be used.

Q. Do the guidelines mention treatment of high triglycerides (TGY) also?

Yes, the guidelines say that if the TGY is above 150 mg/dL and below 500 mg/dl we should use Non HDL as target goal. There is increased risk of heart disease because of remnant cholesterol in triglycerides. One of the common causes of uncontrolled TGY is uncontrolled diabetes and thyroid disease. . Lifestyle modification like absenteeism from alcohol, regular exercise, reducing carbohydrate diet and reducing weight is the first step to reduce Triglyceride. However, if levels are still high and patient is suffering from diabetes or cardiovascular diseases, we recommend to use the statin and non statin drugs. Fish oil (EPA) is also helpful in patients with high triglycerides.

If TGY levels are above 500mg then the aim is to prevent pancreatitis. Low fat diet and aggressive lifestyle modification and drugs like Fenofibrate EPA (fish oil) may be used. In patients with high TGY and DM or fatty liver ,Saroglitazor is useful.

Q. Is genetic cause of lipid disorder more common in India?

Yes, familial hypercholesterolemia, a genetic disorder of lipid metabolism is much more common. Prevalence of FH is more than 15% in younger heart attack patients. Should be treated with statin and non-statin drugs to keep LDL<55 mg/dl.

By cascade screening we can diagnose carriers of this disease in the family (siblings and children of index case) whom we can treat early in asymptomatic stage and prevent premature heart disease.

Elevated Lp(a) (>50 mg/dl) is the most common genetic disorder of lipid metabolism affecting 25% of the general population and is associated with increased heart attack, stroke, heart failure and aortic valve disease. Elevated levels are seen in 35% of heart attack patients and more than 40% in HeFH population. It should be measured atleast once.

Q. How important are lifestyle changes important to manage dyslipidemia?

Yes, exercise and Yoga helps to reduce Triglycerides, increase HDL and make the size of LDL cholesterol larger.

In our guideline we have discussed the role of diet based on the Indian population. Studies have shown that high consumption of carbohydrate and sweets cause more blockage than moderate consumption of saturated fats.

Q. Do the guidelines and goal of lipid are different in children and young adults?

Yes, young heart attack patients (18-40 yrs) do not have any different strategies. Those who are carriers of Heterozygous Familial Hypercholesterolemia (HeFH), and Young adults with risk factors should be treated with lifestyle modification and small to moderate dose of statin to keep the LDL-C below 100mg/dl.

Q. While working through the guidelines, what were the main challenges concluded by the expert group in managing dyslipidemia in India?

A number of challenges exist in society like unidentified dyslipidemia patients, inadequate treatment, non adherence to therapy, physician's inertia, negative media and cost.

The proposed guidelines focus on the dictum 'earlier the better' which calls for early detection, 'lower the better' which calls for keeping LDL low and longer the better which means throughout life and together the better which is about using combined statin and non statin to achieve goal safely and effectively .

These guidelines do not replace the individual clinician judgement that remains the sole in charge in the management of patients.

Q. To conclude, can you sum up the main takeaways you have come across while coming up with these guidelines.

I think a couple of very important points like lipid profile (non fasting), BP measurement and blood sugar should be done at the age of 18 years. One must always look for HeFH in young CVD patients and by cascade screening of family members we can identify carriers and treat them early to prevent premature disease. Treatment with statin and non stain drugs are safe and one should adhere to treatment under guidance of physicians and to ensure that LDL-C goals are attained. Finally, the guidelines suggest that all biochemical laboratories should have a uniform reference range for Lipids.